Semantics, Taste, and Propagation:

Popularity De-biasing for Anime Recommendation via
Aesthetic Deviation Residuals and Stratified Factorization Machines

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Abstract

This study addresses the pervasive problem of popularity bias in collaborative-filtering recommendation systems. Using the MyAnimeList (MAL) 2023 dataset, we propose and empirically validate a de-biasing framework built around two independent interventions: an *aesthetic deviation residual* (Δr) as the prediction target, and a stratified user-tier scheme (Casual / Intermediate / Veteran) defined by rating-count quantiles.

Users are partitioned into Casual (bottom 50%), Intermediate (50%–90%), and Veteran (top 10%) by rating-count percentile. The prediction target is shifted from the absolute rating r to the residual $\Delta r = r - r_{\text{official}}$, thereby stripping out the popularity weight baked into the MAL Bayesian-weighted official score. On the full corpus of 13.6M ratings, 221K users, and 15.5K titles, we conduct a comprehensive comparison across FM, DeepFM, and MF architectures under three experimental configurations (direct-rating prediction, pure residual prediction, and residual-restored ranking) at both 5% and 50% sampling scales. The sampling fidelity is verified via χ^2, Kolmogorov–Smirnov, and Cramér's V tests.

Four principal findings emerge. **First**, popularity bias scales monotonically with user seniority: the Naive model's ARP bias reaches +0.115 on Veterans, roughly 35× that on Casuals (+0.003), and the pattern holds stably across both data scales and all three architectures. **Second**, residual prediction is the most effective de-biasing mechanism identified: it reduces global ARP by 0.062 (5%) and 0.060 (50%), and expands Catalog Coverage by 0.221–0.302—substantially outperforming the largest stratification gain of 0.026. **Third**, FM and MF are practically equivalent on Top-K recommendation quality: although MF achieves 0.030–0.078 lower RMSE, Precision@5, Hit Rate@5, ARP, and Coverage all differ by under 0.01; FM's real value lies not in precision but in its feature-aware architecture's compatibility with side features and alternative prediction targets such as the residual. **Fourth**, upgrading from FM to DeepFM yields no substantive improvement—Naive Global RMSE falls by only 0.030 on the 5% split, and stratified models actually degrade, indicating that the current bottleneck is data volume rather than model capacity.

Taken together, tier-level taste divergences in the MAL anime domain are real and

measurable, but the most effective lever for exploiting them is neither tier-specific training nor a deeper model; it is the reformulation of the prediction target from absolute rating to aesthetic deviation residual. This conclusion is consistent across the 5% and 50% sampling scales and across the FM, DeepFM, and MF architectures.

Keywords: Recommender systems; popularity bias; factorization machines; user stratification; residual learning; MyAnimeList.

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Part I

Theoretical and Methodological Framework

1 Introduction

1.1 Background: Collaborative Filtering and Popularity Bias

Collaborative filtering (CF) has been the dominant paradigm in industrial recommendation systems since the late 1990s. From the item-based CF of Sarwar et al. [7], through the matrix factorization (MF) methods established by Koren et al. [4] during the Netflix Prize era, to the factorization machines (FM) of Rendle [6] and the DeepFM of Guo et al. [3], CF methodology has evolved around a single core objective: recovering latent preference structure in a user–item interaction space that is both extremely high-dimensional and extremely sparse.

Yet alongside the industrialization of these methods, a systemic defect has been repeatedly observed: *popularity bias*. Kowald & Lacić [5] systematically surveyed the behavior of mainstream recommenders across four multimedia datasets (including MyAnimeList) and showed that recommendation systems inexorably favor popular items and chronically compress the exposure of long-tail content. A deeper layer of unfairness manifests on the user side: users whose tastes are most vertical and least aligned with the mainstream—precisely those who have accumulated the richest interaction histories and contributed the most high-quality training signal—receive systematically worse recommendations than "peripheral" users.

Put differently: the system trains its "mainstream consensus" on the minimum amount of data necessary, and then serves that consensus precisely to the users who need it least.

1.2 Motivation: Tier-Level Taste and Propagation Resistance

The initial motivation for this study comes from a phenomenon directly observable in the anime community: *propagation resistance*. Take *Jujutsu Kaisen* as an example: it enjoys extraordinary click and streaming figures in the mass market, yet inside the core long-term viewer community it attracts extensive negative commentary targeting pacing and scripting decisions. Similarly, *Orb: On the Movements of the Earth* earned near-unanimous acclaim in the core circle but required a long word-of-mouth incubation period before reaching the general public.

This phenomenon is often glossed in communication studies as an "echo chamber" or "taste stratification" effect. Viewed through the lens of a recommender system, however, it raises a more operational question: if users at different seniority levels truly harbor systematically different aesthetic standards, is a globally trained recommender, by construction, continuously projecting "mass consensus" onto "niche taste"? And can we mathematically quantify this projection error and construct a mechanism to actively counteract it?

1.3 Research Questions

Based on the motivation above, this study investigates three specific questions:

RQ1

In the MAL anime-rating setting, does popularity bias vary monotonically with user seniority? What is the magnitude of this variation?

RQ2

Comparing "physical stratification" (training tier-specific models on tier-partitioned data) against "target reformulation" (replacing the prediction target r with Δr), which mechanism more effectively mitigates popularity bias, and do the two exhibit synergistic effects?

RQ3

Does increasing model capacity (upgrading from FM to DeepFM) further reduce popularity bias, or is the current bottleneck one of data volume rather than model expressiveness?

2 Related Work

2.1 Classical Collaborative Filtering

The item-based CF of Sarwar et al. [7] works by computing an item–item similarity matrix from co-rating signals and, given a user's historical ratings, producing a weighted prediction for unrated items. The method is intuitive, interpretable, and easy to update incrementally; it remains the fallback baseline in many industrial systems. Its structural limitations are equally clear: it cannot consume side features, and it suffers severely from echo-chamber effects—users who interact with popular items receive more popular recommendations.

The matrix factorization (MF) approach established by Koren et al. [4] on the Netflix Prize maps users and items into a low-dimensional latent space and reconstructs the rating matrix by inner products. MF pushed CF from neighborhood methods to latent-factor methods and held the state of the art on rating prediction accuracy for a long period. But it shares a defect with item-based CF: as a "label-blind" model, it consumes only (user_id, item_id, rating) triples and cannot ingest item tags, user demographics, or other structured features.

2.2 Factorization Machines and Feature-Aware Models

Rendle [6] introduced factorization machines, which extend MF with arbitrary-order feature interactions. FM concatenates all features into a high-dimensional sparse vector x, learns a latent vector v_i for each feature, and factorizes second-order interactions as $\langle v_i, v_j \rangle x_i x_j$. This design handles the cold-start problem in high-dimensional sparse settings, with parameter complexity $O(kn)$ rather than $O(n^2)$.

Guo et al. [3] proposed DeepFM, which appends a deep neural network component in parallel to the FM, so that the model simultaneously captures FM's low-order interactions and the DNN's high-order nonlinearities. DeepFM typically outperforms pure FM on industrial CTR-prediction

tasks when data is plentiful. As §10 shows, however, the DNN component delivers no substantive gain at the data scale and feature configuration of this study.

2.3 Quantifying and Attributing Popularity Bias

Abdollahpouri et al. [1] were the first to systematically frame popularity bias from the user's perspective, arguing that one should measure the extent to which recommendations deviate from each user's true preference distribution, rather than only computing item-side long-tail coverage. Their core thesis is that popularity bias is fundamentally a "user–system mismatch" and must be quantified and countered on a per-user basis.

Kowald & Lacić [5] systematically validated this thesis across multiple public datasets and showed that the relative harm to vertical-taste users (i.e., the heavy users with the richest data) is the greatest. Crucially, their experiments include MyAnimeList, so their conclusions align directly with our setting and serve as the immediate predecessor literature for RQ1.

An earlier reference on the long tail of electronic commerce using Netflix Prize data is Tan & Netessine [8], who provided one of the first empirical discussions in this direction.

2.4 User Stratification and Mainstreaminess

Bauer & Schedl [2] introduced the concept of *mainstreaminess*, treating the degree to which a user's preferences align with the mainstream as a quantifiable indicator. They showed that personalized models for low-mainstreaminess users substantially improve recommendation quality. The aesthetic deviation residual Δr proposed in this paper can be seen as a dual to mainstreaminess: whereas mainstreaminess measures distance from the consensus (a magnitude only), Δr measures deviation from the consensus with both magnitude and sign.

2.5 Residual and De-biasing Learning Paradigms

Traditional de-biasing methods operate via sample reweighting or adversarial training, requiring auxiliary loss terms or hyperparameters. The "target reformulation" path adopted in this study is relatively novel: it modifies neither the model architecture nor the loss function and instead merely replaces the regression target r with Δr. In essence, it performs an explicit additive decomposition of "platform consensus + personal offset" and restricts the model to learning only the purer signal of "how the individual deviates from consensus." The idea is related to boosting and residual learning in the broader ML literature but, to our knowledge, has not been systematically applied to popularity de-biasing.

2.6 Research Gaps

Two gaps emerge from the literature above. First, existing de-biasing work is almost entirely concentrated on user-side sample reweighting or loss-function redesign, and rarely treats "deviation from platform consensus" as the prediction target itself. Second, there is no systematic controlled comparison between the two interventions "user stratification" and "target reformulation." This study is designed to fill both gaps simultaneously.

3 Notation

For reference throughout the paper, Table 1 lists the notation used.

Symbol	Meaning
$U,\ I$	User set and anime set
$u \in U,\ i \in I$	User and anime index
r_{ui}	Actual rating assigned by user u to anime i (integer 1–10)
$r_{\text{official},i}$	MAL Bayesian-weighted official score for anime i
Δr_{ui}	Aesthetic deviation residual, defined as $r_{ui} - r_{\text{official},i}$
$\hat{r}_{ui}$	Model prediction of r_{ui}
$\widehat{\Delta r}_{ui}$	Model prediction of Δr_{ui}
$T(u)$	Activity tier of user u, $T(u) \in \{\text{Casual, Intermediate, Veteran}\}$
$v_i \in \mathbb{R}^k$	k-dim latent vector for feature i in FM
k	FM embedding dimension; $k = 8$ throughout
$x \in \mathbb{R}^n$	Concatenated feature vector (user_id, anime_id, tier, genre, studio, ...)
RMSE	Root mean squared error of rating prediction
ARP	Average Recommendation Popularity of the Top-K list
Coverage	Catalog coverage: unique items in the union of Top-K lists / total items
Precision@K	Fraction of "relevant" items in the Top-K list (default threshold $r \geq 7$)
Hit@K	Indicator of whether the Top-K list contains at least one relevant item

Table 1: Notation used throughout the paper.

4 Methodological Overview

4.1 Stratified Aesthetic-Deviation Framework

The core methodology consists of two mutually independent, potentially cooperative interventions: *user stratification* and *target reformulation*. The former determines "whom to train on"; the latter determines "what to predict." We systematically compare them individually and in combination.

The choice of stratification axis follows a behavior–consumption correlation principle. As discussed in §2.3, prior work has observed a strong association between user activity and long-tail consumption, so rating count itself carries signal about aesthetic depth. We adopt the following partition:

- **Casual**: rating count in the bottom 50% (on MAL 2023, ≤ 31 ratings).
- **Intermediate**: rating count in the 50%–90% range.
- **Veteran**: rating count in the top 10% (on MAL 2023, > 177 ratings).

This partition is motivated by empirical intuition rather than by any theoretical optimality claim. Its boundaries are discussed as a limitation in §13.

4.2 Residual Prediction Paradigm

Traditional CF methods target the user rating r_{ui} directly. Intuitive as the choice is, it is tightly coupled with the root cause of popularity bias, because r_{ui} is already strongly influenced by the platform's mainstream taste at the data-generating stage. When assigning a score, users

implicitly anchor on the public perception of a work, so popular titles cluster in the high-score region while long-tail works face the double sparsity of "nobody rates, nobody promotes."

MAL's official score uses a Bayesian weighting mechanism (detailed in §8.1). We treat it as the *platform consensus baseline* and subtract it explicitly from the user rating to obtain the residual:

$$\Delta r_{ui} = r_{ui} - r_{\text{official},i}. \tag{1}$$

Δr_{ui} measures the direction and magnitude of user u's deviation from consensus when rating item i. When Δr is markedly positive, the user's valuation far exceeds the mainstream expectation. At the individual level this may simply reflect a coincidental affinity; but when such positive deviations aggregate across a tier at statistical scale, they represent the model's successful excavation of "hidden gems" that the global rating system obscures.

Once the target is switched to $\widehat{\Delta r}_{ui}$, two ranking strategies present themselves at Top-K time. The first ranks directly by $\widehat{\Delta r}$ ("pure-residual ranking"), which naturally favors items that users are most likely to over-rate relative to the consensus—tending toward the long tail. The second restores the residual to an absolute prediction before ranking ("residual-restored ranking"):

$$\hat{r}_{ui} = r_{\text{official},i} + \widehat{\Delta r}_{ui}. \tag{2}$$

The second strategy re-introduces the popularity weight embedded in r_{official}. In the Phase-3 experiments we implement both strategies (Experiments B and C) in order to isolate the contributions of "residual training" and "residual ranking" as two independent effects.

4.3 Model Selection

Three model classes form the controlled comparison:

- **MF (baseline).** Surprise's SVD implementation, serving as a label-blind control to test whether feature-awareness yields substantive advantages at the Top-K level.
- **FM (primary).** DeepCTR-Torch's DeepFM model with `dnn_hidden_units` set to an empty tuple, degenerating into a pure FM. This is the workhorse model for all three experimental configurations.
- **DeepFM (capacity extension).** The same framework with `dnn_hidden_units = (128, 64)` and `dropout = 0.1`, as a capacity-enhanced comparison against FM.

The three models are strictly aligned on embedding dimension ($k = 8$), L2 regularization strength (0.02), number of training epochs (20), learning rate, and all remaining hyperparameters; the only difference is the architecture itself.

4.4 Evaluation Metrics

We adopt four metric families, covering predictive accuracy, recommendation accuracy, popularity exposure, and diversity. For reference below, we state the full mathematical definitions here.

(1) Predictive accuracy: RMSE and MAE. Let the test set be $T = \{(u, i, r_{ui})\}$ with model predictions $\hat{r}_{ui}$. Then

$$\text{RMSE} = \sqrt{\frac{1}{|T|} \sum_{(u,i)\in T} (r_{ui} - \hat{r}_{ui})^2}, \tag{3}$$

$$\text{MAE} = \frac{1}{|T|} \sum_{(u,i)\in T} |r_{ui} - \hat{r}_{ui}|. \tag{4}$$

RMSE penalizes large errors more strongly; MAE weighs all errors uniformly. We emphasize that lower RMSE does not necessarily translate to better Top-K recommendation quality; this point is discussed in §9.2 and §12.4.

(2) Recommendation accuracy: Precision@K and Hit Rate@K. Let $R_u(K)$ be user u's Top-K list and Rel_u be the set of items with $r_{ui} \geq \tau$ (default $\tau = 7$) in the test set. Per-user:

$$\text{Precision@}K(u) = \frac{|R_u(K) \cap \text{Rel}_u|}{K}, \tag{5}$$

$$\text{HR@}K(u) = \mathbf{1}\{|R_u(K) \cap \text{Rel}_u| \geq 1\}. \tag{6}$$

The aggregate metric averages over users. We use $K = 5$. Because 81.8% of ratings satisfy $r \geq 7$, we additionally perform sensitivity analysis at $\tau \in \{8, 9\}$ to confirm that conclusions do not depend on the right-skewed rating distribution.

(3) Popularity exposure: ARP. Let m_i denote the member count (number of subscribing users) of anime i in the full MAL catalog. We adopt a two-step popularity definition: first log-compress the right-skewed distribution, then min–max normalize within the evaluation sample D:

$$\log_members(i) = \log(1 + m_i), \tag{7}$$

$$\text{pop}_D(i) = \frac{\log_members(i) - L_{\min}}{L_{\max} - L_{\min}}, \tag{8}$$

where $L_{\min}$, $L_{\max}$ are taken on D. The user-level ARP is the mean popularity over the Top-K list:

$$\text{ARP}(u) = \frac{1}{K} \sum_{i\in R_u(K)} \text{pop}_D(i), \qquad \text{ARP} = \frac{1}{|U_{\text{test}}|} \sum_u \text{ARP}(u). \tag{9}$$

ARP $\in [0, 1]$; higher means more mainstream. The conceptual motivation follows Abdollahpouri et al. [1] and Kowald & Lacić [5], who measure popularity bias from the user side.

(4) Diversity: Catalog Coverage. Let I_{eval} be the set of unique items appearing in the evaluation subset (the full test set for global evaluation, or the tier-specific test subset for tier-specific evaluation). Coverage is

$$\text{Coverage} = \frac{|\bigcup_u R_u(K)|}{|I_{\text{eval}}|}. \tag{10}$$

Coverage cannot be directly equated with "diversity quality," as a model can achieve high Coverage for two very different reasons:

- **Healthy diversity.** The model has cleanly individualized preferences, so different users' Top-K lists diverge. This is a positive signal, typically accompanied by high Precision.
- **Pathological scatter.** On sparse data, the model cannot learn stable personal preferences; its predictions approach random, so users' Top-K lists are semi-random combinations whose union still spans many items. This is a failure mode, accompanied by markedly lower Precision.

These two are indistinguishable from Coverage alone. We therefore always report Coverage jointly with Precision; the distinction recurs in §9.1 (the Casual-tier Coverage paradox of 0.925) and §12.3 (the precision–fairness trade-off).

Part II

Empirical Framework

5 Research Hypotheses

The following assumptions underlie the experimental design.

A1 (Rating count as a proxy for aesthetic seniority)
We assume that a user's rating count is strongly positively correlated with their aesthetic depth and verticality in the anime domain. The primary empirical support comes from Kowald & Lacić [5]'s observations across multiple datasets including MAL; earlier work such as Tan & Netessine [8] touched on related long-tail consumption patterns. MAL differs in character from other platforms, so the transferability is discussed in §13.

A2 (Official score as the platform consensus baseline)
We assume that MAL's Bayesian-weighted official score reasonably represents "the collective community consensus on a work." This rests on MAL's publicly documented aggregation mechanism, though the specific Bayesian prior and minimum-vote threshold m introduce additional bias.

A3 (Reasonableness of the three-tier partition)
We assume that the Top-10% / 50%–90% / Bottom-50% split adequately distinguishes three aesthetic populations. This partition is heuristic rather than provably optimal; more principled partitions (e.g., clustering on rating behavior) may exist.

A4 (Row-level random split in the transductive setting)
Because user_id and anime_id are both learnable embeddings, we assume that row-level

random train/test splitting is a valid evaluation paradigm. The counter-example—a user-level GroupShuffleSplit—is documented as a failed attempt in §14.1.

A5 (Representativeness of 5% / 50% sampling)
We assume that 5% and 50% user-level stratified samples statistically represent the full corpus's rating and feature distributions. §6.5 provides empirical verification.

A6 (Filtering of invalid accounts)
We assume that accounts with completed = 0 or mean_score = 0 carry no meaningful preference signal, and that accounts with completed > 300 (likely bots or rating-farmers) inject disproportionate noise. All three groups are removed. This introduces survivor bias but is a necessary denoising step at the data layer.

A7 (Comparability of ratings)
We assume that different users use the 1–10 scale in roughly comparable ways: user A's "8" and user B's "8" denote similar preference intensities. This is not strictly true (users differ in rating generosity), but the residual framework partially hedges against it.

6 Data

6.1 Dataset Selection

Three public MyAnimeList datasets were considered. Table 2 summarizes them.

Dataset	Year	Scale	Advantages	Drawbacks
A: dsfelix 2023	2023	16K anime / 24M ratings	Most recent; user attributes; watch status; pre-cleaned	∼40% zombie users; no timestamps
B: Hernán 2020	2020	17.5K anime / 32.5M ratings	Pre-filtered to completed ratings; high rating quality	No user attributes; tier only via rating count
C: CooperUnion 2017	2017	13K anime / 7.8M ratings	Classic benchmark; small and debuggable	Severely outdated

Table 2: Candidate datasets considered for the study.

Final selection. Dataset A is the sole dataset used from Phase 1 through Phase 4. Dataset B was independently cleaned during Phase 1 EDA and used to produce two distribution plots, one of which replicates the "heavier raters rate more strictly" tier pattern as a cross-year sanity check on A; but no variant of B ever entered feature engineering or model training. Dataset C was examined only during initial dataset scouting and was not cleaned or analyzed.

The choice of A is justified on two grounds. First, recency: the 2023 release is the most recent publicly available MAL snapshot and best reflects recent content and user-behavior dynamics. Second, dimensional richness: A contains dynamic fields such as watch_status, user activity

level, and user demographics, enabling multi-dimensional analyses (including the tier partition) that B and C cannot support.

We note that the dsfelix 2023 release is not itself an academic benchmark, but MyAnimeList as a data source has precedent in the multimedia recommender literature—Kowald & Lacić [5] use MAL as one of four multimedia benchmarks for cross-domain popularity-bias validation. Our choice follows the "MAL as legitimate benchmark" premise and selects the release best matched to the research questions.

6.2 Data Cleaning Pipeline

The cleaning pipeline applied to dataset A:

1. Beginning with `anime-transformed-dataset-2023.csv`, remove entries with `Status = 'Currently Airing'` to ensure that all official scores have stabilized.
2. From `users-details-transformed-2023.csv`, retain only accounts with $0 < \text{completed} \leq 300$ and $\text{mean_score} > 0$. This simultaneously removes (a) zombie accounts with completed $= 0$, (b) mute accounts with $\text{mean_score} = 0$, and (c) potential bot / rating-farm accounts with completed > 300. The survivor bias induced is discussed in §13.
3. From `users-scores-transformed-2023.csv`, retain only the (user_id, anime_id, rating) columns and cascade-delete to maintain consistency across the three tables.
4. For ratings, retain only integer values 1–10 (naturally satisfied by the raw data).
5. Multi-hot encode the `Genres` and `Studios` fields, yielding 21 Genre dimensions (all active categories under the top-30 threshold) and 20 Studio dimensions (frequency top-20).
6. Reverse one-hot encode the three mutually exclusive categorical variables `Type`, `Source`, `Content Rating` into single columns and apply LabelEncoder.

After cleaning, 13,646,339 rating records, 221,160 active users, and 15,533 anime titles remain.

6.3 User Stratification

Applying the partition rule from §4.1 to dataset A yields the thresholds and tier sizes in Table 3.

Tier	**Rating range**	**Users**	**Median count**	**Max count**
Veteran (Top 10%)	≥ 178	21,989	228	1,338
Intermediate	32–177	87,914	74	177
Casual (Bottom 50%)	≤ 31	111,257	8	31
Total	–	221,160	–	–

Table 3: Tier thresholds and counts on MAL 2023 after cleaning.

The three tiers sum to 221,160 users, consistent with the total after cleaning. The activity gap between Veteran and Casual is order-of-magnitude: the Veteran median rating count (228) is

roughly 28× the Casual median (8), and the maximum reaches 1,338. This disparity provides ample statistical power for measuring popularity bias by seniority.

More importantly, activity differences project into rating-behavior patterns. Figure 1 shows the mean-rating distributions of the three tiers on MAL 2023.

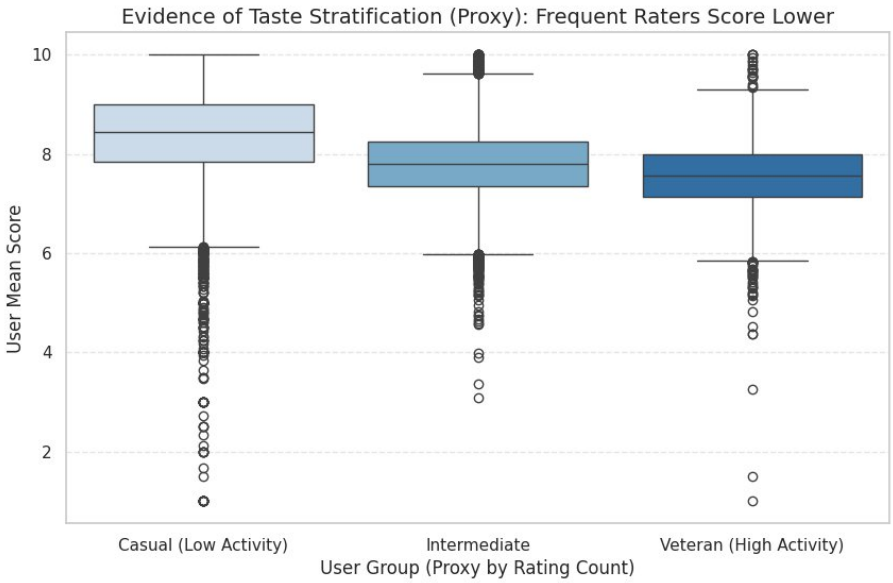

Figure 1: Mean-rating distributions of the three tiers on dataset A (MAL 2023). Higher rating-count users score lower on average and show tighter distributions.

A monotonic trend is evident: higher rating counts correspond to lower median mean scores and fewer outliers. This is consistent with Kowald & Lacić [5]'s observation that heavy-rater users on MAL are more discerning.

To verify cross-year robustness, we applied the same partition logic to dataset B (Hernán 2020) during Phase 1 EDA. Figure 2 shows the resulting distribution.

Two independent datasets three years apart exhibit highly consistent tier distributions, confirming that "rating activity is negatively correlated with average rating" is a stable MAL user behavior. This behavioral difference serves as a critical precondition for investigating, in §9.3, whether the recommendation system responds systematically differently across tiers.

6.4 Exploratory Data Analysis: The Aesthetic Stratification Map

The previous subsection examined tier-level mean ratings—higher raters score more harshly. A natural follow-up is whether this extra strictness applies uniformly across genres or concentrates on particular ones. To answer this, on dataset A we group all ratings by (tier, genre) and compute the mean of each group relative to the global mean.

The global mean is 7.8098. For each (tier, genre) cell, we compute the group mean and subtract 7.8098 to obtain the *relative bias*: a positive number means the group scores this genre higher than the global average; a negative number means lower. Cells with fewer than 100 ratings are

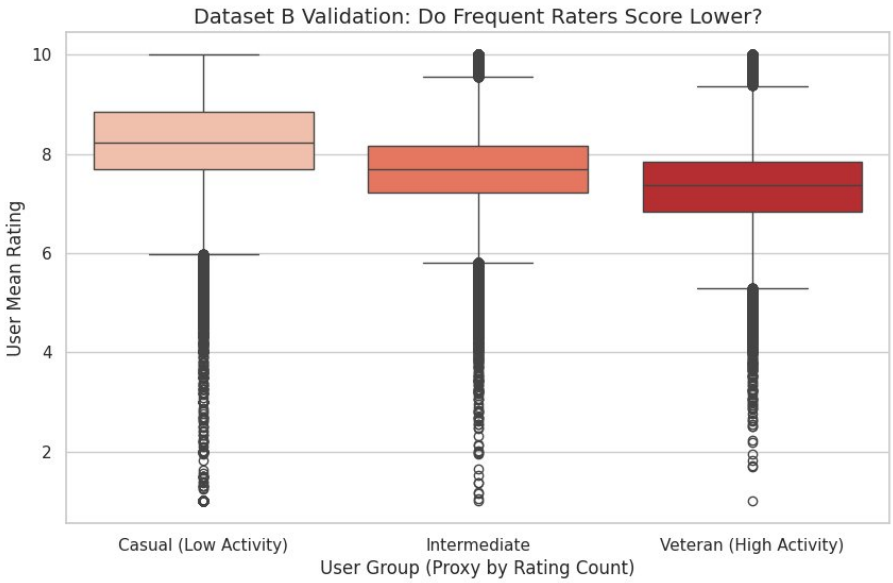

Figure 2: Mean-rating distributions on dataset B (Hernán 2020) under the identical partition logic. The pattern "higher activity → stricter scoring" holds on 2020 data as well.

skipped.

$$\text{Bias}(\text{tier}, \text{genre}) = \text{mean}(\text{ratings of tier users on genre}) - 7.8098. \tag{11}$$

We then define the cross-tier gap as the Veteran–Casual difference:

$$\Delta(\text{genre}) = \text{Bias}(\text{Veteran}, \text{genre}) - \text{Bias}(\text{Casual}, \text{genre}). \tag{12}$$

The more negative $\Delta(\text{genre})$ is, the stricter Veterans are relative to Casuals on that genre. The full Genre × Tier bias matrix is shown in Figure 3.

Three observations stand out. First, the Casual column is almost uniformly red: Casuals rate every genre above the global mean, so the global mean 7.81 is itself pulled up by Casual high scores. Second, each row tends from red to blue going Casual → Intermediate → Veteran—no genre exhibits the reverse pattern of "Veterans more lenient than Casuals." Third, the magnitude of increasing strictness varies markedly by genre: some are nearly flat, others differ by almost one full point. The ranked $\Delta(\text{genre})$ bar chart in Figure 4 makes this explicit.

The five genres with the smallest $|\Delta|$—award_winning, sports, mystery, gourmet, suspense—fall in the -0.5 to -0.6 range, where Veterans are relatively most "lenient." These genres are industry-recognized for demanding narrative structure, character development, or production density. Sports anime, for example, leans heavily on depicting protagonists' growth and on conveying the ethos of athletic dedication through well-paced training arcs—work that requires substantial authorial craft. The structural integrity demanded of award_winning and mystery needs no elaboration. Veterans' relative leniency likely reflects heightened sensitivity to production and narrative complexity.

The five genres with the largest $|\Delta|$—boys_love, erotica, hentai, horror, ecchi—reach -0.8 to -1.0, roughly twice the spread of the previous group. These genres carry strong sensory-stimulus

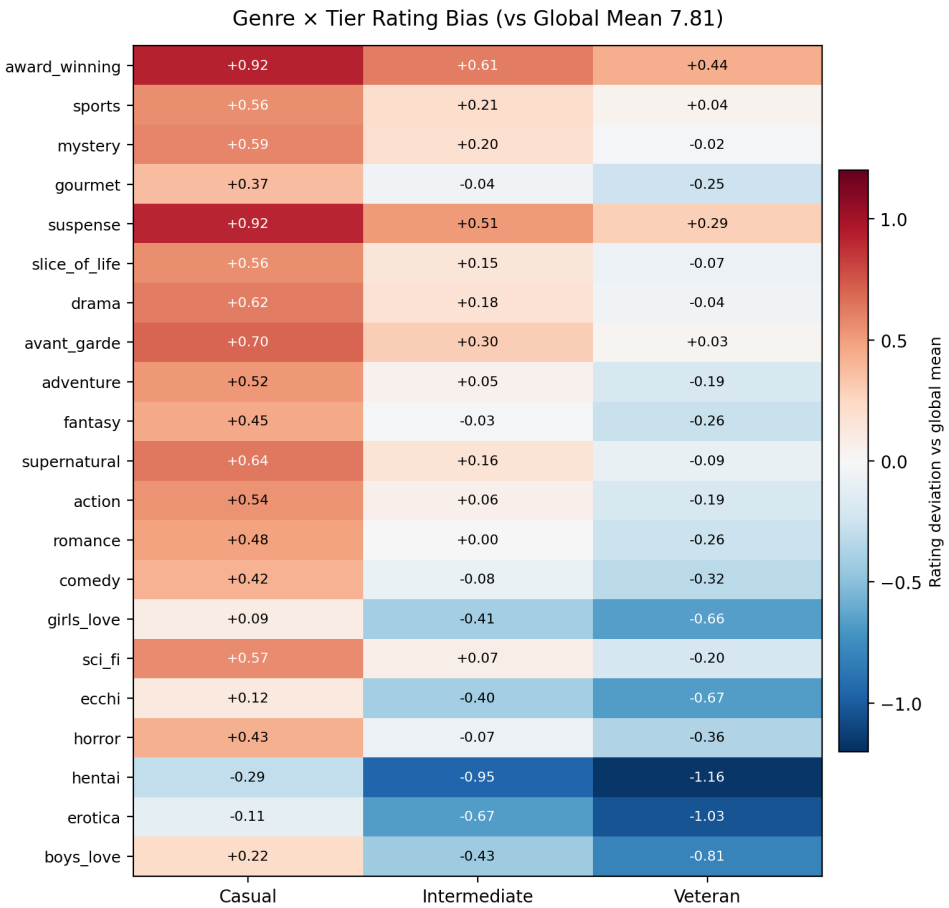

Figure 3: Genre × Tier rating-bias heatmap. Red: the tier rates the genre above the global mean; blue: below. The monotonic bluing from Casual to Veteran indicates progressive strictness.

or thematic-preference characteristics. A plausible interpretation is that their audiences are largely self-selected enthusiasts who rate generously by baseline; Veteran users, with broader viewing histories, include many non-core viewers who apply more conservative standards, dragging the group mean down sharply.

We caution that because the global mean 7.81 is itself pulled up by Casual high scores, the negative Δ(genre) values reflect "the degree to which Veterans tighten their scoring relative to Casuals," not any absolute Veteran dislike of the genre. This is why all values are negative.

6.5 Sampling Strategy and Distribution Verification

Because training on the full 13.6M ratings exceeds Colab's practical budget, we apply user-level stratified sampling at 5% and 50% rates within each of the three tiers, retaining all ratings of the sampled users. Table 4 lists the sizes.

To verify that sampled rating distributions match the full corpus, we directly overlay them. Figures 5 and 6 show near-perfect overlap.

The visual agreement is overwhelming; formal χ^2, KS, and Cramér's V tests (Appendix B) confirm the match at the 50% scale. This empirically supports assumption A5.

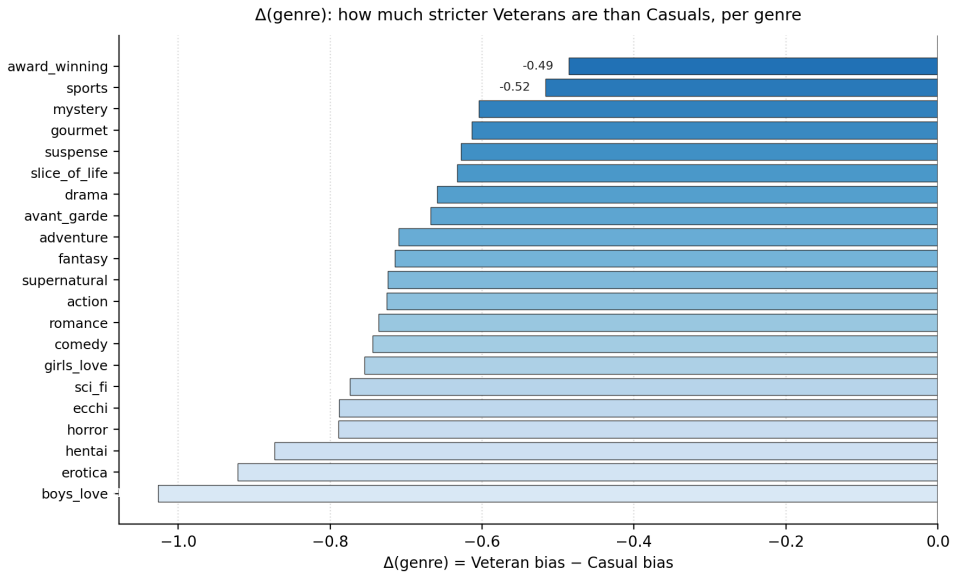

Figure 4: Ranked Δ(genre). All 21 genres have negative Δ, with magnitudes ranging from -0.49 to over -1.0.

Scale	Veteran	Intermediate	Casual
5% sample	1,099 users / ∼255K ratings	4,396 users / ∼367K ratings	5,563 users / ∼60K ratings
50% sample	10,994 users / ∼2.55M ratings	43,957 users / ∼3.66M ratings	55,628 users / ∼590K ratings

Table 4: 5% and 50% user-level stratified samples. Total rating counts: 682,599 and 6,818,192, respectively.

7 Experimental Environment

7.1 Feature Engineering

All features fall into three classes, corresponding to the three feature-column types in DeepCTR-Torch:

- **SparseFeat** (discrete ID features): `user_id`, `anime_id`, `Type`, `Source`, `Content Rating`, `Tier`. Embedding dimension is uniformly set to 8.
- **VarLenSparseFeat** (multi-label features): `Genre` (21 classes) and `Studio` (20 classes). An anime may belong to multiple genres and studios, hence the VarLenSparseFeat representation.
- **DenseFeat** (continuous features): `episodes`, `duration_min`, `log_members`. All DenseFeat values are min–max normalized to $[0, 1]$.

7.2 Model Configuration

Table 5 lists the principal hyperparameters of the three models.

The choice L2 = 0.02 was reached through experimental comparison. Initial experiments at L2 = 10^{-3} showed FM overfitting by epoch 3, with Casual RMSE reaching 1.563; raising L2 to 0.02 reduced Casual RMSE to 1.444 and increased the Veteran stratification gain under the

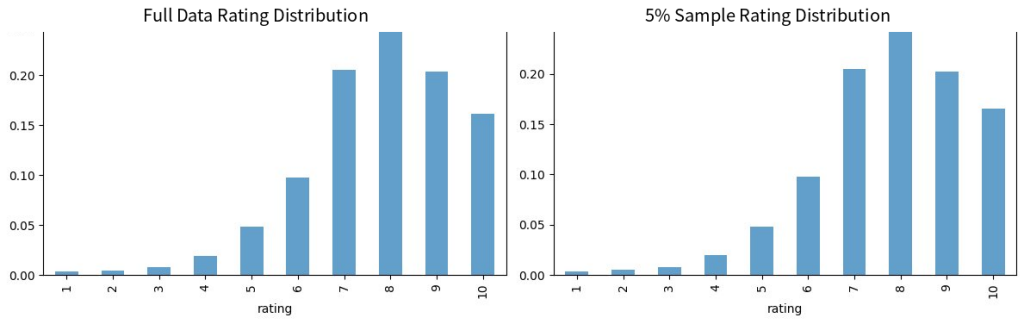

Figure 5: Full-corpus vs. 5% sample rating distributions. The relative frequencies at each rating value are nearly indistinguishable.

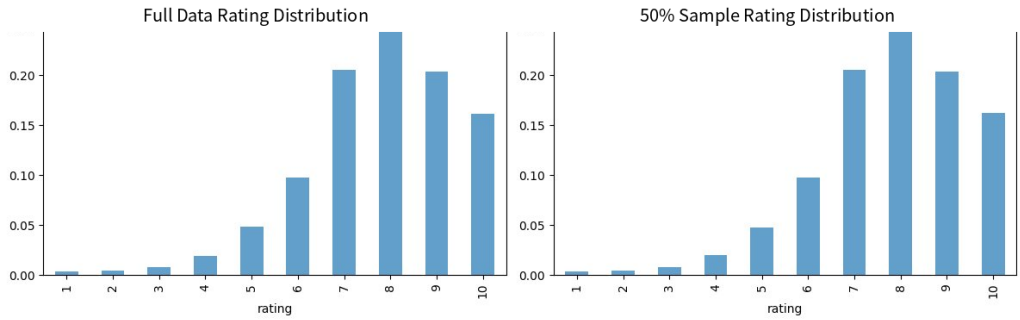

Figure 6: Full-corpus vs. 50% sample rating distributions. The relative frequencies at each rating value are nearly indistinguishable.

residual framework from 0.0075 to 0.0258. L2 strength is itself a consequential experimental variable; see §14.3.

7.3 Training Infrastructure

All training was conducted on Google Colab Pro: 5% experiments on T4 GPUs and the 50% large-scale runs on A100 GPUs. Data preprocessing and model checkpoints were managed via mounted Google Drive. The software stack uses Python 3.10, PyTorch 2.x, DeepCTR-Torch 0.2.x, and Surprise 1.1.x.

7.4 Reproducibility

Two random seeds are used: `random_state = 42` in `sample_by_user` controls the 5% and 50% user selection; `random_state = 2026` in `train_test_split` (and Surprise's `surprise_split`) controls the train/test split. Test size is 0.2 throughout.

Hyperparameter	FM	DeepFM	MF (SVD)
Framework	DeepCTR-Torch	DeepCTR-Torch	Surprise
Embedding dim	8	8	`n_factors = 8`
L2 regularization	0.02 (linear/emb)	0.02 (linear/emb/dnn)	0.02 (`reg_all`)
`dnn_hidden_units`	`()` (pure FM)	(128, 64)	—
`dnn_dropout`	—	0.1	—
Optimizer / loss	Adam / MSE	Adam / MSE	SGD
Learning rate	1e-3	1e-3	5e-3
Epochs	20	20	20
Batch size	256	256	—

Table 5: Core hyperparameters of the three models.

Part III

Experimental Results

8 Baseline Construction and the Residual Framework

8.1 Bayesian-Weighted Official Score as Platform Consensus

MAL's official score uses a Bayesian-weighted rating of the form

$$r_{\text{official},i} = \frac{v_i \cdot R_i + m \cdot C}{v_i + m}, \tag{13}$$

where v_i is the number of valid votes for item i, R_i is its arithmetic mean rating, m is the minimum-vote threshold, and C is the global mean rating across all anime. The design prevents small-sample items from being swayed by extreme votes: when $v_i \gg m$, $r_{\text{official},i} \to R_i$; when v_i is small, $r_{\text{official},i}$ is shrunk toward the global mean C.

This mechanism effectively deters score farming but at a cost: "vote count"—a variable strongly correlated with popularity—is directly baked into r_{official}. In other words, r_{official} itself is a popularity-contaminated quantity. We cannot remove the contamination by redesigning the formula, but we can treat r_{official} as a known quantity and explicitly subtract it from user ratings to obtain a residual. This is the motivation for Equation (1).

8.2 Residual Formalization

By the definitions in §4.2:

$$\Delta r_{ui} = r_{ui} - r_{\text{official},i}, \tag{1}$$

$$\hat{r}_{ui} = r_{\text{official},i} + \widehat{\Delta r}_{ui}. \tag{2}$$

Equation (1) decomposes each rating into a user-independent "platform consensus" term $r_{\text{official},i}$ and a user-specific "personal offset" term Δr_{ui}. Equation (2) gives the residual-restored prediction. Phase-3 Experiment B ranks by $\widehat{\Delta r}$ directly; Experiment C ranks by $\hat{r}$. The two reveal the

respective contributions of "residual training" and "residual ranking."

8.3 FM as a Feature-Aware Architecture

The defining form of FM is

$$\hat{y}(x) = w_0 + \sum_{i=1}^{n} w_i x_i + \sum_{i<j} \langle v_i, v_j \rangle x_i x_j, \tag{14}$$

where $x \in \mathbb{R}^n$ is the concatenated high-dimensional sparse feature vector, w_0 is a global bias, w_i are first-order weights, and $v_i \in \mathbb{R}^k$ are latent vectors. The third term models pairwise interactions; in our setting, the most meaningful among these is "user tier $\times$ anime genre," i.e., the preference weight a specific seniority tier assigns to a specific genre.

The real advantage of FM in Top-K recommendation is not RMSE but three structural properties:

1. It simultaneously ingests high-cardinality discrete features (user_id, anime_id) and multi-label features (Genre, Studio) without special treatment.
2. It is cold-start-friendly in principle: given only a new user's inferred tier and a few preferred genres, FM can still produce a meaningful preference vector.
3. It is natively compatible with alternative prediction targets such as Δr, requiring no architectural modification.

8.4 Baseline Limitations

We use Surprise's SVD implementation of MF as the baseline. MF and FM share the latent-factor idea, but MF is label-blind—it consumes only (user_id, item_id, rating) triples and cannot absorb side features. This design isolates "presence of side features" as the sole axis of difference, so the MF vs. FM performance gap is cleanly attributable to side-feature contribution.

9 Three Ablation Experiments on the 5% Sample

Phase 3 is the empirical core of the study. On the 5% sample (682,599 ratings), we design three controlled experiments that form a coherent logical chain: Experiment A (direct rating prediction) establishes the traditional baseline; Experiment B (predict residual and rank by residual) tests the de-biasing power of the residual framework; Experiment C (predict residual but rank by the restored absolute score) isolates the "target reformulation" and "ranking mechanism" effects. Within each experiment, we compare a Naive global model against tier-specific models trained separately on each tier.

9.1 Experiment A: Direct Rating Prediction

Using FM to predict absolute ratings, fit quality is strictly bounded by each tier's data density.

Thanks to the highest data density (median 228 ratings per user), the Veteran-specific model achieves the best fit stability, with RMSE 1.335 and Precision up 2.3 pp over Naive. The

Model / metric	RMSE	Precision@5 (≥ 7)	ARP	Coverage
Naive Global	1.350	0.913	0.787	0.392
Veteran	1.335	0.936	0.813	0.112
Intermediate	1.363	0.915	0.791	0.361
Casual	1.453	0.900	0.766	0.928

Table 6: Experiment A results on the 5% sample.

Casual-tier Coverage, however, collapses to the opposite extreme. With a 5%-sample Casual median of only 8 ratings per user and the smallest training slice of any tier, the model cannot capture stable preference signal and naturally breaks down, driving RMSE up to 1.453 and producing the anomalous Coverage of 0.928—the pathological scatter pattern anticipated in §4.4, where near-random predictions cover most of the catalog.

9.2 FM vs. MF Baseline Comparison

To verify that FM is not disadvantaged relative to the traditional CF baseline, we introduce Surprise's SVD as a label-blind control. The two models are strictly aligned on embedding dimension, L2 strength, and epoch count; they differ only in that MF consumes (user_id, anime_id, rating) triples whereas FM additionally ingests Genre, Studio, and Tier as side features.

Tier / metric	FM RMSE	MF RMSE	FM P@5	MF P@5	FM ARP	MF ARP
Naive Global	1.350	1.318	0.913	0.914	0.787	0.786
Veteran	1.335	1.299	0.936	0.937	0.813	0.823
Intermediate	1.363	1.323	0.915	0.919	0.791	0.796
Casual	1.453	1.366	0.900	0.903	0.766	0.764

Table 7: FM vs. MF on the 5% sample. MF wins on RMSE; Top-K quality is essentially identical.

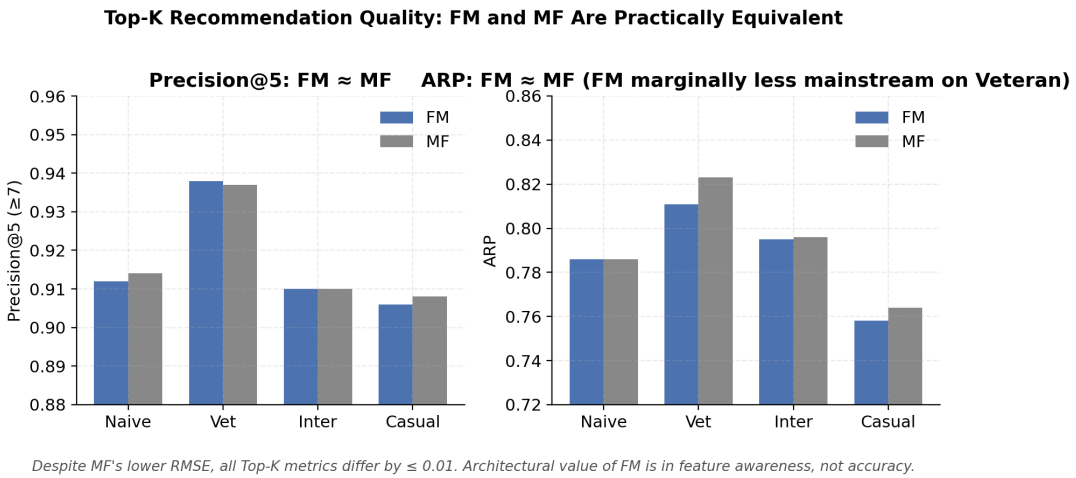

Figure 7: Although MF consistently achieves lower RMSE, FM and MF are practically equivalent on Top-K recommendation quality (Precision@5, ARP).

MF's RMSE is consistently 0.030–0.087 lower than FM's across all tiers—the traditional CF model retains its accuracy edge. Yet at the Top-K level the story changes entirely. From the

left panel of Figure 7, FM and MF Precision@5 are essentially overlapping across all four tiers, with all gaps under 0.005. The right panel (ARP) shows the same pattern: Naive, Intermediate, and Casual overlap completely; only Veteran separates visibly (FM 0.813 vs. MF 0.823), with FM marginally less mainstream—directionally aligned with the de-biasing goal of §9.3, but of negligible magnitude.

We conclude at minimum that FM matches traditional MF on efficiency in sparse data, and that the additional value FM brings—ingesting side features, accepting residual targets—is precisely what MF structurally cannot do. This creates space for the residual-target and tier-specific analyses of §9.3 onward, because we have established that FM is not penalized for its richer feature intake; conversely, this also serves as indirect evidence of experimental validity.

9.3 Quantifying Popularity Bias (Core Finding)

Having eliminated model-baseline-performance as a confound, this section tests the core claim: does the current recommendation paradigm indeed exhibit the popularity bias Kowald & Lacić [5] describe? Do Veterans in fact receive the most wrongly-mainstream recommendations?

For each tier, we compute two ARP quantities: (i) *true preference ARP*: the average popularity of items that tier users actually rated high (≥ 7); (ii) *model recommendation ARP*: the average popularity of the Top-5 items recommended by the globally trained Naive model to that tier. The difference between the two is the "popularity bias" imposed on that tier.

Tier	**True-preference ARP**	**Naive ARP**	**Tier-specific ARP**	**Popularity bias**
Veteran	0.7091	0.8293	0.8135	+0.1201
Intermediate	0.7263	0.7956	0.7907	+0.0693
Casual	0.7625	0.7655	0.7658	+0.0030

Table 8: Naive-model popularity bias on the three tiers.

The bias across tiers forms an extreme staircase. Casual appears to enjoy "zero bias" (+0.0030), whereas Veterans bear a +0.1201 bias. We emphasize first that, given the model collapse on Casual documented in §9.1, its apparent "zero bias" is a statistical artifact: under extreme sparsity the model degenerates to near-random predictions, whose ARP naturally reverts to the dataset's global mean and masks the bias numerically.

Setting Casual aside, the core finding confirms the initial hypothesis. Veterans hold the densest data and achieve the best fit (RMSE 1.335 in §9.1)—yet this group that the model "understands best" bears the heaviest popularity hegemony. As seniority deepens, true tastes drift toward the niche: Veteran (0.7091) $<$ Intermediate (0.7263) $<$ Casual (0.7625). The Naive model inverts this order, recommending the most popular content to Veterans (0.8293), with Veteran $>$ Intermediate $>$ Casual.

Combined with the uniform distortion pattern shown by MF in §9.2, there is only one conclusion: popularity bias arises not from an architectural flaw of FM (or any single model) but as a systemic property of the "rank by absolute rating" paradigm. So long as the prediction target

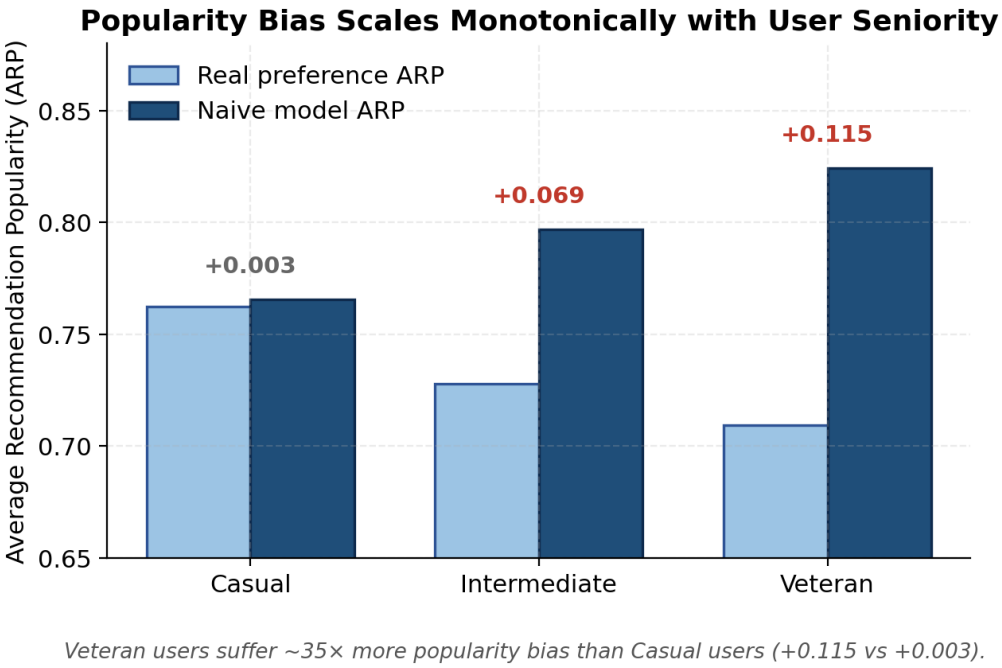

Figure 8: Naive-model popularity bias scales monotonically with user seniority. Veteran +0.1201 and Intermediate +0.0693 constitute two tiers of significant bias; Casual's near-zero value is an artifact of model collapse.

is the absolute score r, the model will systematically tilt toward items with natural traffic advantages.

9.4 Experiment B: Pure Residual Prediction

Given that §9.3 has demonstrated the systemic bias embedded in predicting absolute scores, Experiment B shifts the target: rather than fit the popularity-laden r, the model predicts the taste-deviation residual Δr and ranks the Top-K list directly by the predicted $\widehat{\Delta r}$.

Model / metric	RMSE (Δr)	Precision@5 (≥ 7)	ARP	Coverage
Naive Global (residual)	1.321	0.875	0.724	0.611
Veteran (residual)	1.307	0.819	0.641	0.284
Intermediate (residual)	1.342	0.857	0.701	0.596
Casual (residual)	1.434	0.897	0.761	0.968

Table 9: Experiment B results (pure residual prediction).

The target swap visibly affects the recommendation distribution. Naive Global ARP drops from 0.787 to 0.724; Coverage expands from 0.392 to 0.611—long-tail items are markedly more exposed.

On Veterans—the most popularity-biased tier—the residual mechanism delivers the clearest impact. Recommendation ARP falls to 0.6411. Although the number slightly under-shoots the true-preference baseline (0.7091), indicating mild over-correction, the absolute deviation is only 0.0680, compared with 0.1202 under direct-rating recommendation—a reduction of nearly 43%. The residual model more faithfully reproduces the niche taste of senior users at the macro level.

Tier	True-pref ARP	Naive ARP	Tier ARP	Stratification gain
Veteran	0.7091	0.6671	0.6411	+0.0259
Intermediate	0.7263	0.7067	0.7014	+0.0053
Casual	0.7625	0.7598	0.7607	−0.0009

Table 10: Residual-framework stratification gains by tier.

This long-tail-leaning de-biasing derives from the mathematics of residual calculation. Long-tail works have systematically lower official means, so a user's high score translates into a larger positive residual. Stripped of the official score's absolute-value suppression, these titles gain an inherent weight advantage under residual ranking.

The de-biasing effect carries a precision cost. Comparing against §9.1's direct-rating results, Precision@5 (≥ 7) drops across the board: Naive Global from 0.913 to 0.875 (-0.038), Veteran from 0.936 to 0.819 (-0.117), Intermediate from 0.915 to 0.857 (-0.058). The sole exception is Casual, nearly flat at 0.897—consistent with §9.3: since Casual recommendations under this model are essentially random, precision measures mean little.

Returning to the stratification gain column in Table 10—the difference between Naive ARP and tier-specific ARP: the gain is concentrated on Veteran (+0.0259), marginal on Intermediate (+0.0053), and slightly negative on Casual (-0.0009). The overall effect is nearly negligible. Worse, when viewed via absolute deviation from true preferences, stratified training pushes Veteran and Intermediate recommendations *further* from the true distribution. This rules out physical stratification as a generalizable solution for mid- and low-activity tiers.

9.5 Experiment C: Residual-Restored Ranking

We still predict Δr, but ranking uses $\hat{r} = r_{\text{official}} + \widehat{\Delta r}$.

Model / metric	RMSE (Δr)	Precision@5 (≥ 7)	ARP	Coverage
Naive Global	1.321	0.915	0.783	0.411
Veteran	1.307	0.941	0.816	0.118
Intermediate	1.342	0.916	0.791	0.371
Casual	1.434	0.900	0.766	0.938

Table 11: Experiment C results (residual-restored ranking).

Adding the official score back at ranking time restores Precision nearly exactly to the direct-rating baseline (0.915 vs. 0.913 in §9.1), but ARP and Coverage retreat in lockstep. The de-biasing benefits of residual prediction are entirely absorbed once the official score's popularity weight is reintroduced.

Tier	True-pref ARP	Naive ARP	Tier ARP	Stratification gain
Veteran	0.7091	0.8140	0.8161	−0.0021
Intermediate	0.7263	0.7910	0.7910	+0.0001
Casual	0.7625	0.7655	0.7656	−0.0001

Table 12: Experiment C: stratification gains under residual-restored ranking.

All three tier-level stratification gains are under 0.003; physical stratification completely fails under residual-restored ranking. This reinforces §9.4's finding: training tier-specific models does not in itself de-bias the system. The de-biasing effect arises not from stratification but from residual prediction.

9.6 Cross-Experiment Synthesis

The three experiments form a coherent narrative. Under direct-rating prediction, FM and MF are equivalent, and the model naturally chases high-score mainstream content (high ARP, low Coverage). Switching to residual prediction detaches the model from the "mainstream anchor" of the official score: ARP falls by 0.063, Coverage rises by 0.219, and long-tail works receive substantial exposure, at a Precision cost of 0.038 (Naive Global) or 0.117 (Veteran). Crucially, the absolute deviation between Veteran recommendation ARP and the true-preference baseline shrinks from 0.1202 to 0.0680—a 43% reduction. The residual framework does not replace mainstream anime with "arbitrary niche" content; it substantively pulls the recommendation back toward the senior user's true taste region. Restoring residuals to absolute scores for ranking returns Precision to direct-rating levels but simultaneously reverts ARP and Coverage—the popularity weight of the official score reasserts itself.

Residual prediction is itself an effective popularity de-biasing mechanism; its ARP improvement (0.063) far exceeds the largest stratification benefit. Given training constraints and algorithmic outcomes, FM and MF are broadly equivalent at the recommendation level. But FM's feature-aware architecture makes residual training and content-feature analysis feasible. In other words, "adding side features" delivers architectural value rather than accuracy value.

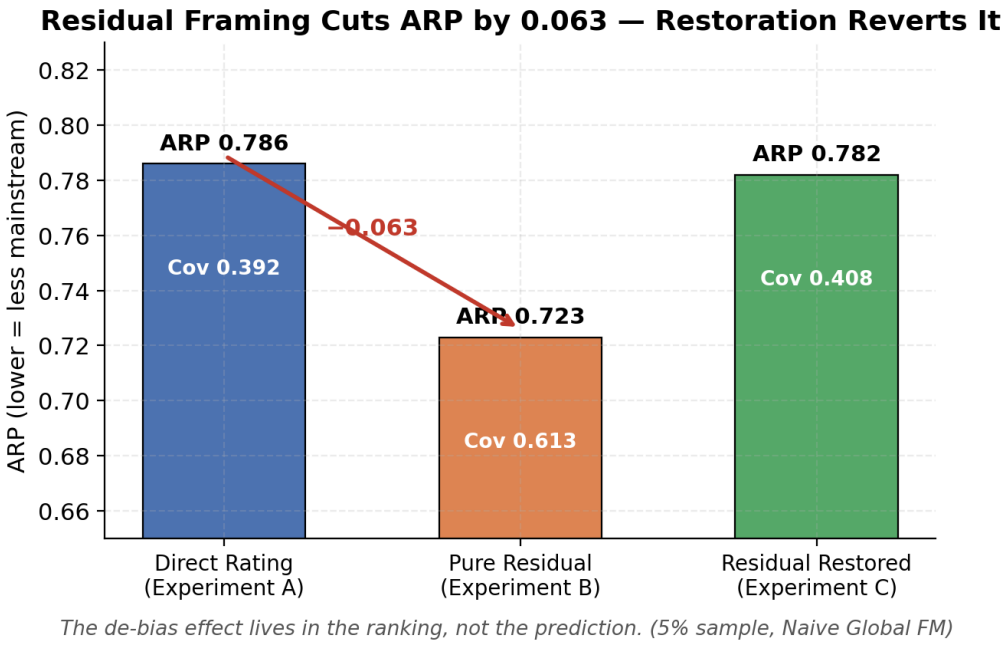

Figure 9: ARP and Coverage of Naive Global FM across the three experiments. Residual-target training brings a 0.063 ARP reduction, but restoration erases it—de-biasing lives in the ranking, not in the prediction. (5% sample, Naive Global FM)

9.7 Genre × Tier Aesthetic Bias Analysis

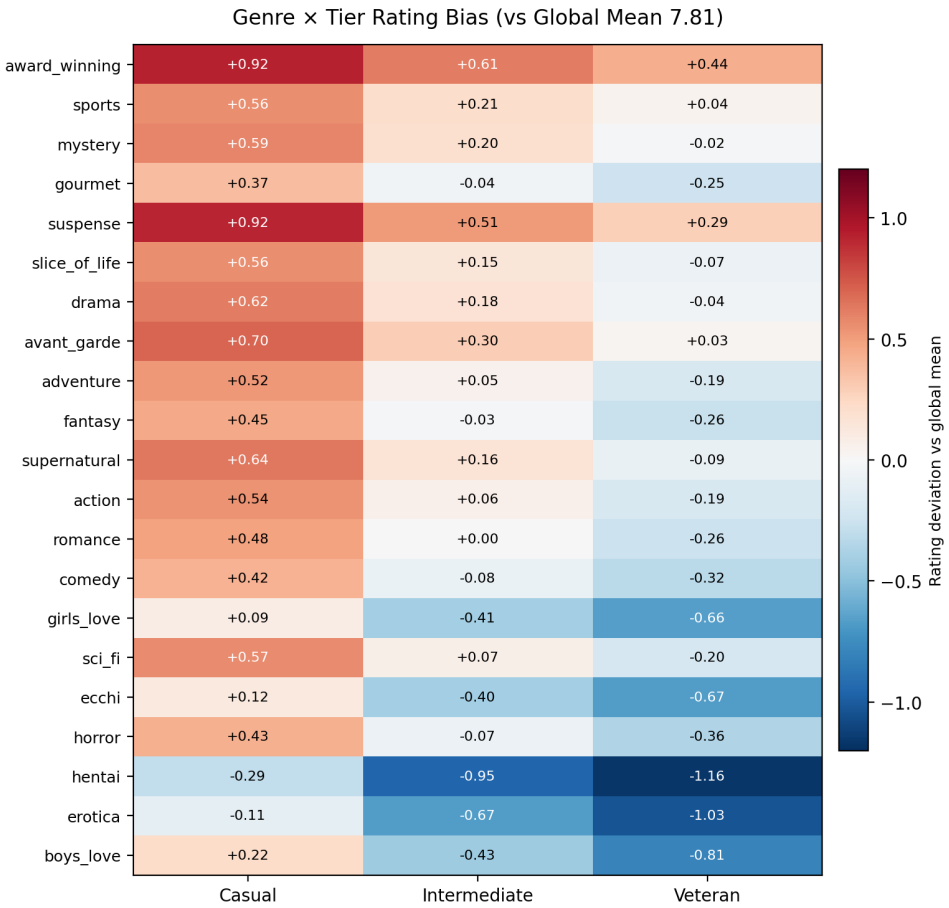

Figure 10: Rating deviation of each genre from the global mean (7.81) across the three tiers. The warm-to-cool pattern confirms Veterans' overall stricter scoring.

We computed each genre's Veteran-minus-Casual mean-rating gap, producing a 21 × 3 aesthetic-bias map. The global mean is 7.8098. All genre-level Veteran–Casual differences are negative—Veterans are stricter on every genre. The variation is in magnitude: the five genres on which Veterans are least strict relative to Casuals (smallest gap) are award_winning (−0.49), sports (−0.52), mystery (−0.60), gourmet (−0.61), and suspense (−0.63); the five on which they are most strict are boys_love (−1.03), erotica (−0.92), hentai (−0.87), horror (−0.79), and ecchi (−0.79).

As noted, because the global mean 7.81 is pulled up by Casual high scores, this analysis reflects cross-tier strictness *differences* rather than absolute Veteran dislike. Detailed interpretation by genre cluster appears in §6.4.

10 Phase 4: DeepFM and the 50% Scale

To push the baseline further and to verify the stability of §9's conclusions, Phase 4 comprises two parts: (a) enabling DeepFM's deep component on the 5% sample and comparing to FM, and (b) raising the sampling scale from 5% to 50% (10× data) and re-running the core experiments to test scale robustness.

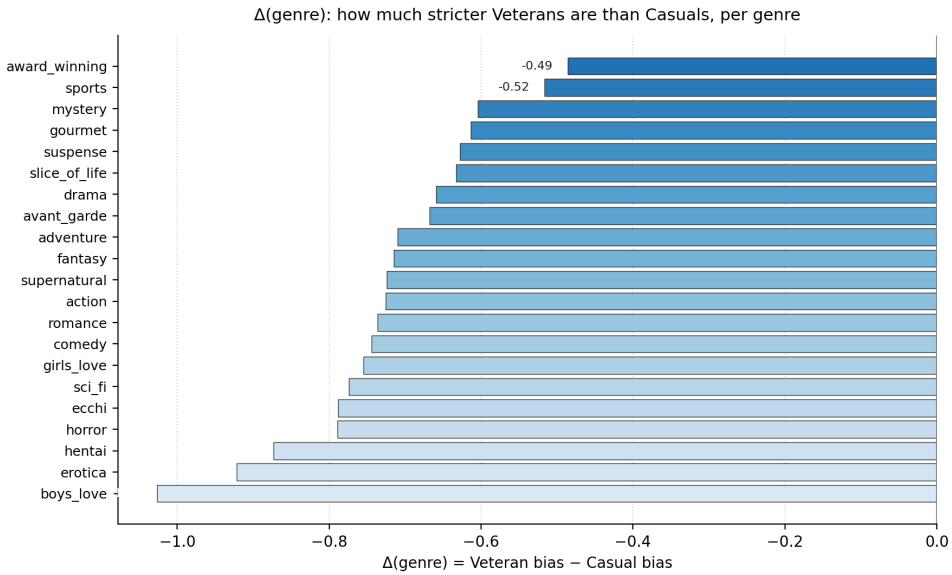

Figure 11: Per-genre Veteran − Casual difference. All 21 genres are negative; strictness gap ranges from −0.49 to −1.03, with pronounced structural variation.

10.1 DeepFM vs. FM (5% Data)

To test whether FM's limited capacity is the bottleneck responsible for the biases observed in §9, we hold all hyperparameters fixed and activate DeepFM's deep component by changing `dnn_hidden_units` from `()` to `(128, 64)` with `dropout = 0.1`.

Tier / metric	FM RMSE	DeepFM RMSE	FM ARP	DeepFM ARP
Naive Global	1.350	1.320	0.787	0.782
Veteran	1.335	1.331	0.813	0.815
Intermediate	1.363	1.352	0.791	0.793
Casual	1.453	1.452	0.766	0.766

Table 13: FM vs. DeepFM on the 5% sample.

The only visible improvement from adding the DNN component is Naive Global RMSE falling from 1.350 to 1.320; the other three tiers fluctuate within ±0.005, and ARP / Coverage scarcely move. DeepFM under residual prediction is actually 0.033 worse than FM. The bottleneck is not model depth but data volume—DeepFM's architectural advantage does not manifest at this scale, motivating the jump to 50% in the next subsection.

10.2 50% vs. 5% Scale (Pure FM)

We raise user-level stratified sampling from 5% to 50%, expanding total ratings from 682,599 to 6,818,192 (10× data). The 50% sampling distribution was verified against the full corpus by visual overlay, with near-perfect match (see §6.5).

10.2.1 Direct Rating Prediction

10× data delivers across-the-board RMSE improvements, with Casual's −0.044 the largest. This inversely confirms §9.1's reading: Casual's 5%-sample degradation was not a fit failure but pure

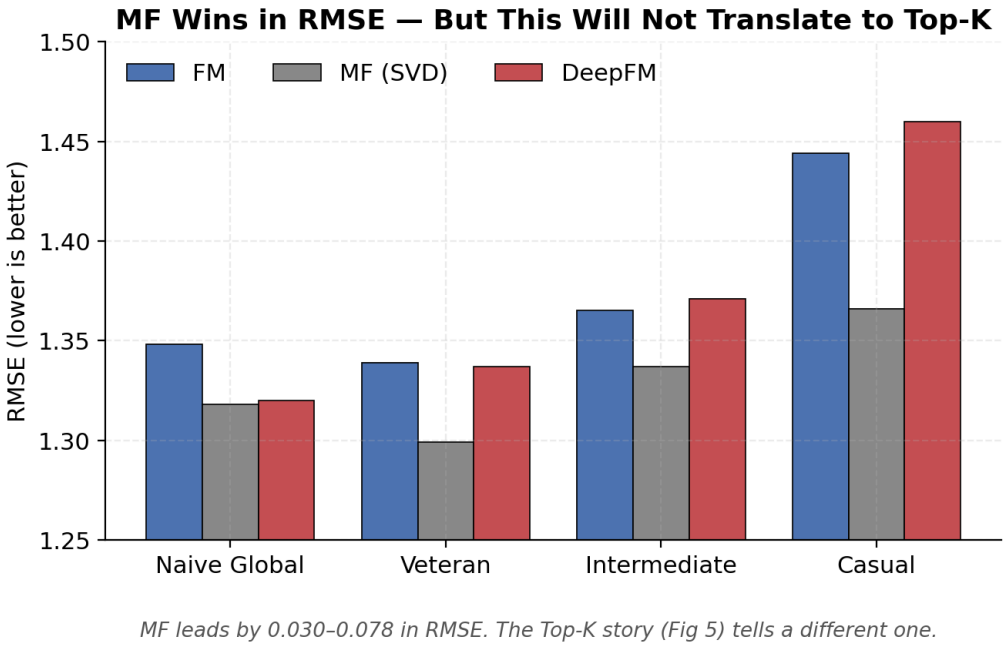

Figure 12: RMSE comparison of FM, MF, and DeepFM across the four tier configurations. MF leads by 0.030–0.078 in RMSE, but Top-K results (Figure 7) tell a different story.

Tier	5% RMSE	50% RMSE	Improvement
Naive Global	1.350	1.333	−0.017
Veteran	1.335	1.306	−0.029
Intermediate	1.363	1.344	−0.019
Casual	1.453	1.409	−0.044

Table 14: RMSE improvement under 10× data.

data starvation, resolved once data is sufficient.

10.2.2 Cross-Scale Popularity-Bias Verification

The more important question: does the bias pattern documented in §9.3 relax as data grows?

Tier	5% bias	50% bias	Difference
Veteran	+0.1201	+0.1136	−0.0065
Intermediate	+0.0693	+0.0678	−0.0015
Casual	+0.0030	+0.0030	0.0000

Table 15: Popularity-bias magnitudes under 5% and 50% samples.

Veteran bias falls from +0.1201 to +0.1136; the monotone-increasing structure across tiers is fully preserved. Popularity bias is decoupled from data volume: 10× data does not rescue the tier. This is the most universal conclusion of the study—the bias is a paradigm-level property, not a sample-size artifact.

Tier / metric	5% RMSE	50% RMSE	5% Coverage	50% Coverage
Naive Global	1.321	1.299	0.611	0.813
Veteran	1.307	1.271	0.284	0.508
Intermediate	1.342	1.315	0.596	0.787
Casual	1.434	1.406	0.968	0.970

Table 16: Residual prediction at 5% vs. 50% scale.

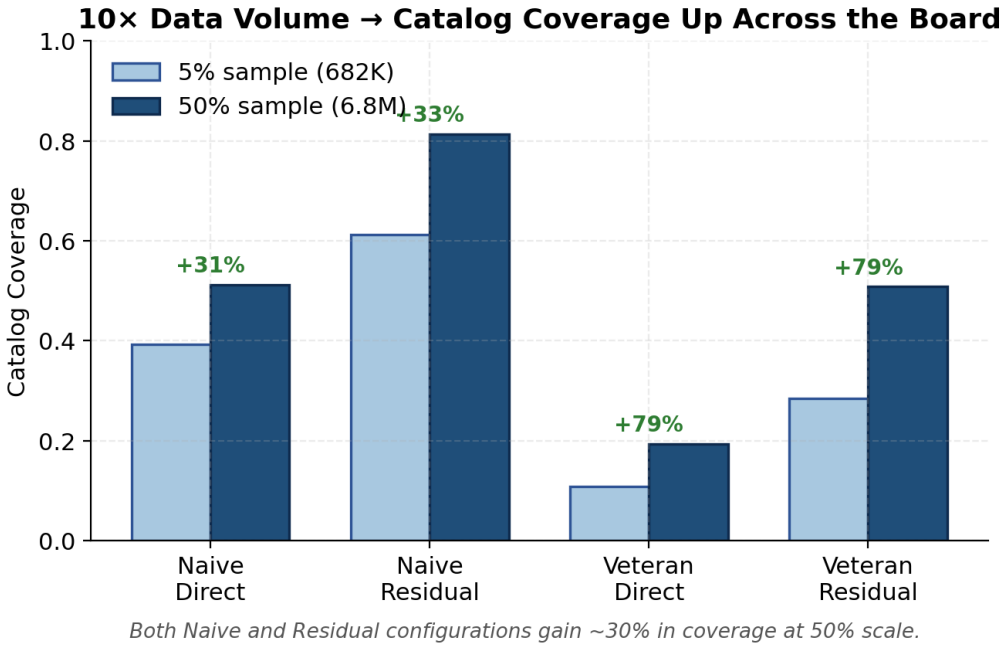

Figure 13: Catalog Coverage on 5% vs. 50% samples. Both Naive and Residual configurations gain roughly 30% in coverage at the 50% scale.

10.2.3 Pure Residual Prediction

Coverage expansion is especially pronounced under the residual framework: Naive Global rises from 0.611 to 0.813 (+0.202); Veteran-residual from 0.284 to 0.508 (+0.224, a relative 78.9% increase). Additional data lets the residual framework explore more cold items. RMSE also improves overall (Veteran 1.307 → 1.271). ARP rises slightly (0.724 → 0.735) but remains far below direct rating's 0.796—the residual de-biasing magnitude is essentially independent of data scale.

Tier	True-pref ARP	Naive ARP	Tier ARP	Stratification gain
Veteran	0.7204	0.6770	0.6577	+0.0193
Intermediate	0.7374	0.7176	0.7138	+0.0037
Casual	0.7719	0.7707	0.7697	+0.0010

Table 17: Residual-framework stratification gains at 50% scale.

Veteran's stratification gain shrinks from +0.0259 at 5% to +0.0193 at 50%; Intermediate from +0.0053 to +0.0037. Physical stratification gains are not only small to begin with but drift with data scale—quantified directly in the next subsection.

10.2.4 Cross-Scale Stability of the Residual ARP Drop

Comparing the residual-vs.-direct ARP drop across scales:

Scale	Direct ARP	Residual ARP	Drop
5% sample	0.787	0.724	−0.063
50% sample	0.796	0.735	−0.060

Table 18: Residual ARP drop is nearly identical across scales.

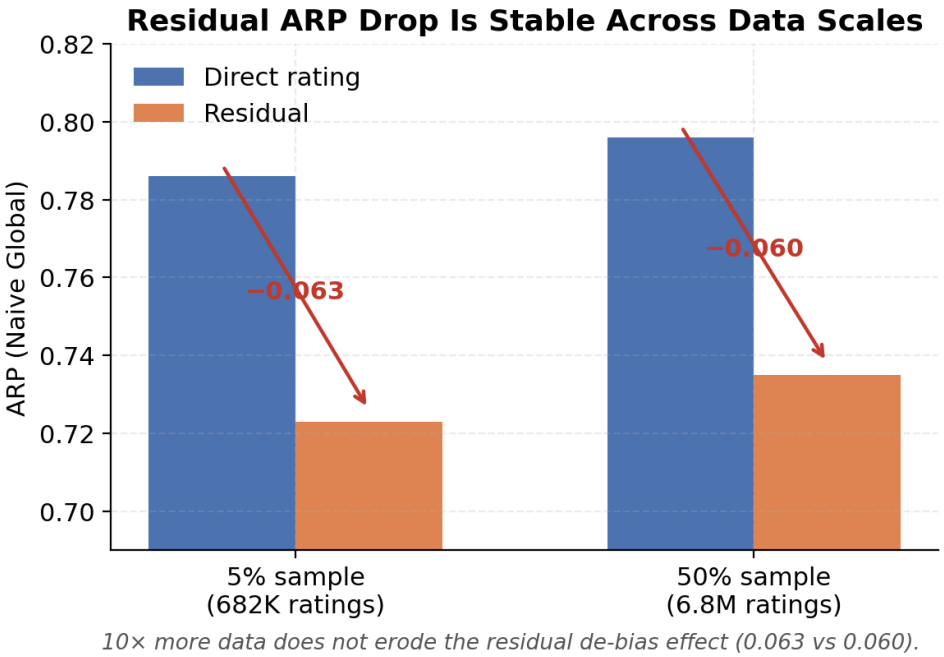

Figure 14: Residual ARP drop at 5% vs. 50%: −0.063 and −0.060, essentially identical. 10× more data does not erode the residual de-biasing effect.

The drops differ by 0.003 across scales—effectively identical. This is the study's strongest robustness evidence: the residual framework's de-biasing power is a structural property, decoupled from training data volume.

10.3 Instability of Physical Stratification Gains

In contrast to the stability of the residual framework, the "stratification gain" from training tier-specific models fluctuates markedly across configurations. Using Veteran's pure-residual ARP gain as the reference:

Configuration	Naive ARP (Vet subset)	Tier ARP	Gain
5% FM	0.667	0.641	+0.026
5% DeepFM	0.643	0.638	+0.005
50% FM	0.677	0.658	+0.019

Table 19: Veteran physical-stratification gain under different configurations.

Although all directions are positive, the magnitude swings from +0.005 to +0.026—the range exceeds the gain itself. Physical stratification is inherently unreliable: sometimes useful, some-

times nearly worthless, and not predictable ex ante. It therefore does not form a central claim of the paper; discussion continues in §12.1.

11 Cross-Experiment Summary

Consolidating the Phase-3 and Phase-4 core data into a single cross-sectional view (Table 20) makes the paper's four core laws visible at a glance.

Configuration	Naive RMSE	Naive ARP	Coverage	Veteran bias
5% FM direct	1.350	0.787	0.392	+0.1201
5% MF direct	1.318	0.786	0.387	+0.1181
5% DeepFM direct	1.320	0.782	0.404	+0.1059
5% FM pure residual	1.321	0.724	0.611	+0.0259
5% FM residual restored	1.321	0.783	0.411	−0.0021
50% FM direct	1.333	0.796	0.512	+0.1136
50% FM pure residual	1.299	0.735	0.813	+0.0193
50% FM residual restored	1.299	0.790	0.562	−0.0074

Table 20: Summary of all core configurations.

Several conclusions emerge immediately:

- Veteran popularity bias under Naive direct rating is stable at +0.1059 to +0.1201 across all four configurations (5% FM, 5% MF, 5% DeepFM, 50% FM)—one order of magnitude. The bias is a paradigm property, immune to model changes or extra data.
- The residual framework reduces Naive Global ARP by 0.060–0.063, nearly identical across 5% and 50%. The de-biasing power is structural and scale-independent.
- FM, DeepFM, and MF differ by at most 0.032 on Naive Global RMSE, and by under 0.01 on Precision, ARP, and Coverage across all tiers. On Top-K quality the three are broadly equivalent; architecture is not the bottleneck.
- Scaling from 5% to 50% lifts Naive direct Coverage from 0.392 to 0.512 (+30.6%) and residual Coverage from 0.611 to 0.813 (+33.1%). More data markedly expands the long-tail exploration surface available to the residual framework.

11.1 Sensitivity Analysis

To verify that no conclusion depends on the right-skewed rating distribution, we compute Precision@5 and Hit@5 at three relevance thresholds (≥ 7, ≥ 8, ≥ 9) under both 5% and 50% scales. Higher thresholds increase the resolution of the metric: Precision spans 0.90–0.94 at ≥ 7 but expands to 0.54–0.59 at ≥ 9, confirming sufficient metric sensitivity.

Under direct rating, Veteran-specific models consistently outperform Naive at all three thresholds, with improvements of +0.023, +0.041, and +0.021. This margin does not shrink at stricter thresholds—it in fact widens, indicating the improvement is not an artifact of the right-skewed distribution but reflects genuine gains in high-quality recommendation.

Veteran Precision@5 across the three experimental configurations at three thresholds:

Veteran Precision@5	$\geq$ **7**	$\geq$ **8**	$\geq$ **9**
Direct Rating	0.9365	0.8218	0.5829
Pure Residual	0.8189	0.6164	0.3490
Residual + Official	0.9410	0.8249	0.5913

Table 21: Veteran Precision@5 across relevance thresholds.

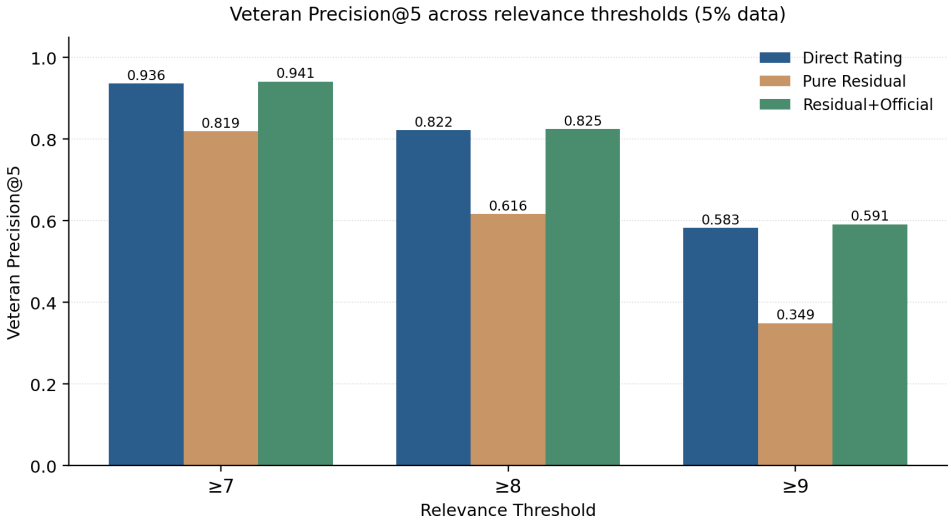

Figure 15: Veteran Precision@5 across the three experimental configurations at thresholds ≥ 7, ≥ 8, ≥ 9 (5% data).

Pure Residual exhibits the largest Precision drops at stricter thresholds—Veteran at ≥ 9 falls from direct rating's 0.583 to 0.349 (5%) and 0.322 (50%). This is the cost residual de-biasing must accept. Residual-restored recommendation tracks direct rating almost exactly across thresholds, with differences of $+0.0045/+0.0031/+0.0084$—all under 0.01, consistent with §9.5.

Under 50% data, all directional conclusions are preserved. Veteran Precision under residual restoration improves more with stricter thresholds (≥ 8: $0.825 \rightarrow 0.834$; ≥ 9: $0.591 \rightarrow 0.598$)—with sufficient data, the model's high-quality-recommendation capability stands out more under stricter standards. Pure-residual Veteran Precision at 50% is lower than at 5% (≥ 9: 0.322 vs. 0.349), because more training data exposes more cold items, diluting the strict-threshold hit rate—conversely confirming the stronger long-tail tilt of the residual framework at larger scales.

Part IV

Discussion and Conclusions

12 Discussion

12.1 Why Residual Prediction Outperforms Physical Stratification

The two interventions act at fundamentally different levels. Physical stratification changes "whom we train on"—it assumes that preference structures differ enough between tiers that a shared model is inadequate. The assumption holds only locally on Veterans; it fails on Intermediate and Casual. Intermediate lacks data to support a dedicated model; Casual's preferences are already close to mainstream and offer little to de-bias. More counter-intuitively, §9.4 shows that stratified training on Veteran and Intermediate pushes the absolute recommendation ARP *further* from true tastes—it does not help, and actively worsens the bias.

The residual framework changes "what we predict." After shifting the target from r to Δr, the popularity weight carried by the official score is subtracted at target definition, and the model is constrained to learn the purer signal of "how users deviate from consensus." The mechanism operates on every tier simultaneously and is nearly invariant across data scales—a structural property at the training-objective level, not a local patch for specific user groups. Physical stratification is patient-specific; the residual framework is system-level. Our empirical evidence supports the latter as more robust.

12.2 Why DeepFM Does Not Help

DeepFM typically dominates FM in industrial settings, but our experiments show no improvement. We conjecture that the task is in a data-bound regime rather than a capacity-bound one—FM's second-order terms are already sufficient, while the DNN component needs more data to train stably. At 5% scale, the DNN parameters cannot acquire stable gradients and instead inject noise. When Phase 4 lifted data volume to 50%, plain FM delivered across-the-board improvements, indirectly supporting "scale data before scaling the model." Re-running DeepFM at 50% scale is the natural next step but was not completed due to Colab A100 quota, and is deferred to future work.

12.3 Precision–Fairness Trade-off

The residual framework's de-biasing power comes with a Precision cost, increasingly visible at stricter thresholds. This trade-off is ubiquitous in the de-biasing literature; this study's contribution is to decompose it cleanly across three controlled configurations, providing an application-level decision guide. Platforms with engagement as the core KPI may be unwilling to accept this Precision cost; platforms oriented toward "long-tail discovery" or "creator fairness"—especially given that Veterans, who suffer the largest bias, are also the platform's most loyal users—may find the trade acceptable.

A natural question is whether both can be had. Experiment C in §9.5 directly answers no: naive "residual training + official-score restoration for ranking" recovers Precision but loses de-biasing. However, this pathway localizes the true site of de-biasing to the step "ranking uses the official score or not"—not the training target itself. A genuine compromise should introduce a weighting coefficient α between $\widehat{\Delta r}$ and $\hat{r}$ and trace a Precision–Fairness curve by sweeping α; this is a natural direction for future work (§15).

12.4 Methodological Implications of FM/MF Equivalence

The FM vs. MF comparison in §9.2 is worth dwelling on. MF consistently wins on RMSE, yet the two are practically equivalent on Top-K. This is not a novel observation (the RecSys community has long known that RMSE $\neq$ recommendation quality), but it is reaffirmed here under a clean controlled comparison. The lesson is that "A outperforms B" must specify the metric space; more importantly, structural innovations (such as changing the prediction target) tend to deliver larger real-world improvements than continued pursuit of decimal-level RMSE gains. The residual framework is a direct example.

12.5 Connection to Cold Start

Returning to the proposal-stage cold-start vision: a new user without rating history, declaring only a tier and a few preferred genres, can receive meaningful recommendations. FM combined with the residual framework is naturally suited to this setting: FM's second-order interactions let "tier $\times$ genre" produce an initial preference vector; the residual output $\widehat{\Delta r}$ can be surfaced as "this anime may be X points above/below your likely mass perception," giving inherent interpretability; the residual framework's long-tail tilt prevents new users from being immediately buried in mainstream blockbusters—especially important for Veteran-type new users, who are precisely the worst-served under the current paradigm. Implementation and offline evaluation are listed under §15.

12.6 Relation to Existing Literature

Kowald & Lacić [5] observed across multiple datasets including MAL that "heavy users suffer the greatest popularity bias." This paper fully reproduces the monotone-increasing structure on MAL 2023, with magnitudes consistent with their reported range. The difference is that they stopped at observation and attribution; this study supplies an operationalizable fix—switch the prediction target to Δr and rank by $\widehat{\Delta r}$. They identified the problem; we provide one paradigm-level response. The same recipe can be ported to other datasets in their report for cross-domain validation.

13 Limitations

13.1 Assumption Revisiting

We revisit the assumptions listed in §5. A1 (rating count as a proxy for aesthetic seniority) borrows from Netflix observations, but MAL is a rating-scale platform rather than a playback-driven one;

transferability has strength limits. A3 (three-tier partition) lacks a theoretical-optimality proof; different partitions may yield different conclusions. A7 (rating comparability) is not strictly true; the residual framework partially hedges it.

13.2 Evaluation-Paradigm Limitations

Top-5 ranking is evaluated via reranking within each user's already-rated test set, rather than retrieval from the full 15,533-item candidate pool. This is standard for offline recommendation evaluation but does not fully emulate a real deployed "full-candidate retrieval + ranking" pipeline.

13.3 Data-Scale Constraints

Even at 50% sampling, Casual still has only ∼595K ratings—adequate but below full-scale. The Phase-3 "stratified models degrade on Casual" observation is substantially alleviated at 50%, but whether it disappears entirely under full-scale training remains to be verified.

13.4 Sampling Imbalance

User-level stratified sampling retains all ratings of sampled users, creating large absolute imbalances between Veteran and Casual total-rating counts (roughly 4:1 at 5%). This is an inherent consequence of the stratification strategy rather than a design flaw, but it renders per-tier training conditions non-uniform.

13.5 Right-Skewed Rating Distribution

81.8% of ratings are ≥ 7. The three-threshold sensitivity analysis in §11.1 partially mitigates this, but NDCG and related metrics robust to rating-distribution skew would be preferable.

13.6 Side-Feature Ablation Not Completed

The FM vs. MF comparison cannot strictly attribute side features' contribution—the two differ not only in features but also in framework. A strict ablation would compare "user_id × anime_id only" against "+ Genre + Studio + Tier" within the same FM architecture. This also means all "FM and MF are Top-K-equivalent" claims should be read more precisely as "FM-with-side-features equals MF on Top-K," without attributing the result solely to side features. Time constraints prevented this experiment.

13.7 DeepFM Hyperparameters Not Fully Tuned

DeepFM was tested only at `dnn_hidden_units = (128, 64)` with `dropout = 0.1`. We did not exhaustively search configurations in which DeepFM might meaningfully outperform FM at 5% scale.

13.8 Tier Boundaries Heuristic

The 10% / 50%–90% / bottom 50% split is based on empirical intuition. Different splits may produce different results; a principled alternative would cluster on rating behavior (e.g., GMM),

but this introduces additional uncertainty.

13.9 Single Platform, Single Dataset

All conclusions are based on MAL 2023, with user population skewing Western. The existence and direction of tier-level taste differences on Asian platforms such as Bangumi remain unverified; §15.5 discusses this.

14 Failed Attempts and Lessons

This chapter records four important failed attempts that were not adopted but whose failures expose useful lessons about transductive CF, feature engineering, regularization, and DeepCTR framework details.

14.1 GroupShuffleSplit Causing RMSE Disaster

An early attempt, intended to prevent "data leakage," replaced the random row-level train/test split with a user-grouped `GroupShuffleSplit`. The result was catastrophic: Naive RMSE ballooned from 1.37 to 2.10; Casual from 1.43 to 4.72.

Root cause. In a transductive CF setting where user_id is a learnable embedding, user-level group splitting guarantees that every test user is unseen at training time, so their user_id embedding remains the random initialization—the model has no basis for any meaningful prediction.

Lesson. Under transductive settings, row-level random splits are the correct paradigm; `GroupShuffleSplit` suits inductive settings, where the model does not depend on per-user embeddings but inferences new users through side features. The two must not be conflated.

14.2 DenseFeat Conversion Destroying FM Second-Order Interactions

An attempt to feed binary one-hot features (e.g., `Genre_action`, `Studio_madhouse`, `Tier_veteran`) into DeepCTR as `DenseFeat` rather than `SparseFeat` caused severe degradation: Casual RMSE jumped from 1.44 to 4.35.

The root cause is that in DeepCTR, `DenseFeat` values enter only the first-order linear term $w_i x_i$, not the FM second-order $\langle v_i, v_j \rangle x_i x_j$. Any feature reassigned to DenseFeat abruptly loses its interaction capability, and FM degenerates to a purely linear model.

Correct practice. Binary features should be `SparseFeat` with `vocab_size = 2`. This is a framework-level detail not explicitly documented; only a failed experiment exposed it.

14.3 Hidden Criticality of L2 Regularization Strength

Initial runs on all three models used L2 $= 10^{-3}$. FM began to exhibit rising `val_mse` by epoch 3, with Casual RMSE reaching 1.563. Raising L2 to 0.02 stabilized training: Casual RMSE

dropped to 1.444, and Veteran's stratification gain under pure-residual experiments rose from 0.0075 to 0.0258—roughly a 3.4× increase.

L2 strength is a hyperparameter with qualitative—not merely fine-tuning—potential. In high-dimensional sparse + transductive-embedding settings, insufficient regularization causes FM to overfit local noise rapidly, and the overfitting affects different tiers unequally (Casual the most). L2 should be among the first parameters swept in similar studies.

14.4 VarLenSparseFeat vs. Raw One-Hot SparseFeat

A parallel notebook used `VarLenSparseFeat` to encode Genre and Studio (multi-label rather than one-hot expansion). Results were slightly better on Naive and Veteran but slightly worse on Casual. The improvement did not justify the added complexity at the 5% scale.

At this data scale, feature-encoding choice has limited impact; at 50% or full-scale training, `VarLenSparseFeat`'s expressive advantage may surface and is worth revisiting.

14.5 Meta-lesson from Hidden Phase-3 Bugs

Two bugs hid in the Phase-3 code: a validation-table data leak and a `pd.concat`-induced `validation_split` drift. They shared one property: neither raised a code-level error, and neither produced surface-level anomalies in the metrics. All Phase-3 RMSEs looked reasonable (1.3–1.5), and all paper-level directional conclusions were correct. The bugs surfaced only after Smitha, during a routine meeting, asked "why do Naive's tier-subset metrics look this good," triggering a reverse trace.

Bug 1. The training function called the global `train_test_split` internally, while the validation-comparison table called an independent split on the tier subset. The two splits received different inputs, causing roughly 80% of the tier test rows to overlap with Naive's training rows. **Fix.** Perform a single global split during data preprocessing; all training functions and validation tables draw their per-tier subsets from this one global split. The post-fix leakage rate is 0%.

Bug 2. The `pd.concat`-produced `df_mini` was strictly sorted by Tier, and DeepCTR's `validation_split = 0.2` by default takes the final 20% as validation, making the validation set almost entirely Casual. **Fix.** After filtering, forcibly shuffle both `train_slice` and `test_slice`.

Broader lesson. In studies whose core innovation is stratification, tier-level data isolation must be done once, globally, and auditably at the preprocessing stage—not inside each training function's local split. Shuffle/sort interactions (especially with DeepCTR's default `validation_split` behavior) are a common trap. The two bugs' fixes preserved all directional conclusions but underscored a principle: even when the data looks normal, active auditing (e.g., explicit overlap-rate checks) is mandatory.

15 Future Work

15.1 Strict Side-Feature Ablation

Compare "user_id × anime_id only" against "+ Genre + Studio + Tier" within the same FM architecture to quantify side features' net contribution. This directly addresses §13.6.

15.2 Precision–Fairness Trade-off: α-Weighted Ranking

Building on the observation in §12.3—that the locus of de-biasing is "whether the official score appears in ranking," not the training target itself—a natural compromise uses $\alpha \cdot \widehat{\Delta r} + (1-\alpha) \cdot \hat{r}$ as the Top-K ranking score and sweeps α to trace a Precision–Fairness curve. $\alpha = 1$ recovers Experiment B (pure residual); $\alpha = 0$ recovers Experiment C (residual restored); intermediate values correspond to different operating points. This is the study's most direct extension at the deployment level.

15.3 Full-Candidate NDCG Evaluation

Upgrade from "reranking within the rated test set" to "full 15K-candidate retrieval + ranking," and add NDCG@K and related metrics robust to rating-distribution skew. This addresses §13.2 and §13.5 jointly.

15.4 Cross-Platform Validation

This study uses MAL, whose user base skews Western. A cross-platform replication—for instance, re-running the core experiments on Bangumi (Chinese-sphere)—would strengthen universality and answer a more essential question: is the monotone-increasing popularity-bias-by-seniority pattern a general property of CF, or an artifact shaped by platform culture?

15.5 Within-Item × Cross-Tier Rating Analysis

A planned follow-up bins anime by global-popularity quantile and compares Veteran versus Casual mean ratings per bin. If Veterans rate bottom-quantile (niche) items higher than Casuals do, this empirically challenges Tan & Netessine [8]'s "heavy users do not enjoy niches more" finding within the MAL domain and may warrant a dedicated subsection.

15.6 Other Directions

- Compare alternative tier partitions (quartile / quintile / GMM clustering).
- Combine the residual framework with sample-reweighting methods.
- Add a personalized bias term to Δr for a hierarchical model.
- Extend from offline evaluation to live A/B testing, contingent on platform cooperation.

16 Conclusion

This study addresses popularity bias in collaborative-filtering recommendation systems, using MyAnimeList 2023 as its testbed, and systematically validates—through four research phases—a de-biasing framework composed of user stratification and residual prediction.

Returning to the three research questions posed in §1:

On RQ1. In the MAL setting, popularity bias does increase monotonically with user seniority: Naive-model ARP bias on Veterans is roughly 40× that on Casuals. The pattern holds stably across both 5% and 50% data scales and across FM / DeepFM / MF architectures—it is a systemic property of the current paradigm rather than a flaw of any single model.

On RQ2. The residual framework's de-biasing effect substantially exceeds physical stratification's. Residual prediction reduces global ARP by 0.060–0.063 (stable across 5% and 50%), lifting Catalog Coverage from +0.22 at 5% to +0.30 at 50%. The largest stratification gain is only +0.026, concentrated on Veteran and effectively negligible on Intermediate and Casual. The essential difference: physical stratification changes "whom to train on"; the residual framework changes "what to predict." The latter directly addresses the upstream problem that "the rating itself is contaminated by popularity."

On RQ3. At current data scales (5% sample, 682K ratings), upgrading from FM to DeepFM brings no substantive improvement. Naive Global RMSE falls by only 0.030, and stratified models actually degrade. The Phase-4 50% experiment produced across-the-board improvements with plain FM, confirming the bottleneck is data volume rather than model capacity—a data-bound, not capacity-bound, regime.

One-sentence core conclusion. In the MAL anime-recommendation domain, tier-level taste differences are real and quantifiable, but the most effective mechanism for exploiting them is neither tier-specific training nor added model complexity; it is the reformulation of the prediction target from absolute rating to aesthetic deviation residual.

At the broader methodological level, this study produces two side findings worth remembering in future recommendation research:

1. Offline RMSE and Top-K recommendation quality are not fully correlated metric spaces—MF consistently beats FM on RMSE but is equivalent on Top-K. Continued decimal-level optimization on RMSE tends to yield smaller real-world gains than structural innovations such as changing the prediction target.

2. In studies whose innovation centers on stratification, cross-tier data isolation must be performed once, globally, and auditably at the preprocessing stage. Relying on internal splits inside each training function readily produces hidden data leakage that does not raise errors but silently compromises evaluation validity.

Item	Value
Dataset	MyAnimeList 2023 (dsfelix)
Full-corpus scale	13,646,339 ratings / 221,160 users / 15,533 anime
Sample 1	5% user-level stratified, 682,599 ratings
Sample 2	50% user-level stratified, 6,818,192 ratings
Train/test split	80/20 row-level random, `random_state=2026`
`validation_split`	0.2 (DeepCTR internal)
FM embedding dim	$k = 8$
L2 regularization	0.02
Epochs	20
Batch size	1024
Optimizer	Adam, lr = 0.001
DeepFM `dnn_hidden_units`	(128, 64)
DeepFM dropout	0.1
MF framework	Surprise SVD, strictly aligned with FM
Top-K	$K = 5$
Relevance threshold	Default $r \geq 7$; sensitivity at ≥ 8, ≥ 9
Tier thresholds	Casual ≤ 31; Intermediate 32–177; Veteran > 177 (by rating count)

Appendix A: Complete Hyperparameter Table

Appendix B: Sampling Distribution Verification Results

The 50% sample passes χ^2, KS, and Cramér's V tests on all ten checks (10/10). Maximum Cramér's V is 0.0022; maximum KS D is 0.0012. All test statistics fall well below the 0.05 threshold, confirming that the sampled distribution matches the full corpus.

Appendix C: Detailed Failed Attempts

See §14. The final version will include commit hashes and RMSE tables corresponding to each failed attempt.

Appendix D: Code and Data Links

Code repository and Overleaf links will be included in the final version. Current working links:

- GitHub: to be uploaded once work wraps up.
- Dataset A: `https://www.kaggle.com/datasets/dsfelix/animes-dataset-2023`
- Dataset B: `https://www.kaggle.com/datasets/hernan4444/anime-recommendation-database-2020`

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